

Inference and Proofs

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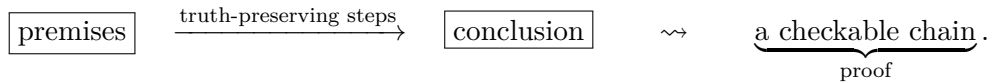
COMP2711 Discrete Mathematical Tools
Rosen, Sections 1.6–1.8

1 Proofs as checkable arguments

A calculation can look persuasive and still contain an illegal step. A conclusion can be true and still fail to follow from the stated premises. This lecture develops a way to make the difference visible.

MUST UNDERSTAND **Proof.** A proof is a valid argument from accepted starting statements that establishes a mathematical claim. It is a certificate whose starting points, local moves, and final claim can all be checked.

The same structure appears at three scales:



We will build that chain in a fixed order. First decide what it means for a whole argument to be valid. Then learn a small collection of trustworthy local moves. Finally use the logical shape of a mathematical claim to choose a proof method.

By the end of the lecture, you should be able to:

- validate an argument or refute it with a countermodel;
- combine propositional and quantified inference rules while respecting their side conditions;
- write complete direct, contraposition, contradiction, case, existence, and uniqueness proofs; and
- choose a proof method from the form of the claim and audit the finished proof.

Term	Working meaning
Premise	a starting statement the argument is allowed to use
Conclusion	the statement the argument claims follows
Argument	premises offered in support of a conclusion
Inference rule	a reusable local step that preserves truth
Proof	the complete checkable certificate just defined

2 Arguments and validity

An **argument** is a sequence of propositions. All but the last are the **premises**; the last is the **conclusion**. We write

$$\frac{P_1 \\ P_2 \\ \vdots \\ P_n}{\therefore C}$$

and ask whether the premises force C .

MUST UNDERSTAND **Valid argument.** An argument is valid exactly when there is no truth assignment on which all premises are true and the conclusion is false. Equivalently,

$$(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow C \text{ is a tautology.}$$

Read the formula as: “If all the premises hold, then the conclusion holds.”

Validity concerns the connection between premises and conclusion, not whether a premise happens to be true in the current situation. The form

$$P \rightarrow Q, \quad P \quad \therefore Q$$

is valid even after substituting a false proposition for P . To use a valid argument to establish a true conclusion, we also need true premises.

Invalid arguments

MUST UNDERSTAND **Countermodel.** A countermodel is one allowed case that makes every premise true and the conclusion false. One row is enough to prove invalidity.

For example, consider

$$P \rightarrow Q, \quad Q \quad \therefore P.$$

Set $P = \text{F}$ and $Q = \text{T}$. Then $P \rightarrow Q$ and Q are true, but P is false. The argument is invalid. This fallacy is often called *affirming the consequent*; the name matters less than seeing the bad direction.

Similarly,

$$P \rightarrow Q, \quad \neg P \quad \therefore \neg Q$$

is invalid. For a concrete reading, let P mean “the vehicle is a car” and Q mean “the vehicle has wheels,” with the rule $P \rightarrow Q$: if the vehicle is a car, then it has wheels. A bicycle gives $P = \text{F}$ and $Q = \text{T}$. Thus having wheels does not prove that the vehicle is a car, and not being a car does not prove that it has no wheels.

To refute validity, do not build an entire truth table unless needed. Search directly for the only kind of row that matters: all premises true and conclusion false.

3 Propositional inference

MUST UNDERSTAND **Rule of inference.** A rule of inference is a valid argument form that can be reused inside larger arguments. Each rule is backed by a tautology. For instance, modus ponens is valid because

$$(P \wedge (P \rightarrow Q)) \rightarrow Q$$

is a tautology.

It is more useful to organize the rules by their job than to memorize one long list.

Implication rules

Rule	Form	Operational reading
Modus ponens	$P, P \rightarrow Q \therefore Q$	Start the implication: P holds, so move to Q .
Modus tollens	$\neg Q, P \rightarrow Q \therefore \neg P$	If Q is false, then P cannot be true.
Hypothetical syllogism	$P \rightarrow Q, Q \rightarrow R \therefore P \rightarrow R$	Chain two guarantees.

Modus tollens follows the contrapositive direction. It is not the invalid pattern “ $\neg P$, therefore $\neg Q$.”

Basic inference rules

Rule	Form	Operational reading
Simplification	$P \wedge Q \therefore P$	Use either known part of a conjunction.
Conjunction	$P, Q \therefore P \wedge Q$	Combine two facts already established.
Addition	$P \therefore P \vee Q$	A true alternative keeps a disjunction true.
Disjunctive syllogism	$P \vee Q, \neg P \therefore Q$	Eliminate one of two alternatives.

Resolution

The **resolution** rule is

$$P \vee Q, \quad \neg P \vee R \quad \therefore \quad Q \vee R.$$

If P is true, the second premise forces R . If P is false, the first premise forces Q . Either way, $Q \vee R$ holds. Resolution is especially useful when formulas are written as disjunctions of propositions or their negations.

3.1 Worked examples

Bus-delay example

Let

D = “the bus is delayed,”

C = “the student misses the connection,”

L = “the student arrives late.”

The journey has two given travel rules:

$D \rightarrow C$ If the bus is delayed, the student misses the connection.

$C \rightarrow L$ If the student misses the connection, the student arrives late.

The observed premise is

$\neg L$ The student did not arrive late.

Formally, all three statements are premises of the argument; the labels distinguish their roles in the story.

Question. Is the bus delayed? Determine whether $D = \text{T}$ or $D = \text{F}$.

Then the proof is locally checkable:

Step	Statement	Reason
1	$D \rightarrow C$	given travel rule
2	$C \rightarrow L$	given travel rule
3	$\neg L$	observed premise
4	$\neg C$	modus tollens, 2 and 3
5	$\neg D$	modus tollens, 1 and 4

Thus $D = \text{F}$: the bus was not delayed. We could instead chain steps 1 and 2 to obtain $D \rightarrow L$, then use modus tollens with step 3. Both arguments are valid. A good proof usually takes the shorter route that remains easy to read.

Case analysis and resolution

Show that $(P \wedge Q) \vee R$ and $R \rightarrow S$ imply $P \vee S$.

The first premise gives two exhaustive cases. They may overlap; proof by cases only requires that every possibility be covered.

1. If $P \wedge Q$ is true, then P is true. Therefore $P \vee S$ is true.
2. If R is true, then S is true because $R \rightarrow S$. Therefore $P \vee S$ is true.

The conclusion holds in both cases, so the argument is valid.

Resolution compresses the same reasoning into symbols. Rewrite $R \rightarrow S$ as $\neg R \vee S$, then apply resolution:

$$(P \wedge Q) \vee R, \quad \neg R \vee S \quad \therefore \quad (P \wedge Q) \vee S.$$

The last formula implies $P \vee S$: its left alternative gives P , and its right alternative gives S . The case proof is usually clearer to a human reader; resolution records the same split as a reusable formal rule.

4 Quantified inference

MUST UNDERSTAND **Quantified inference.** Predicate logic permits two controlled moves between quantified statements and statements about particular objects:

- **Instantiation** removes a quantifier to obtain a statement about one object; for example, $\forall x P(x) \therefore P(a)$.
- **Generalization** adds a quantifier to a statement about one object; for example, $P(a) \therefore \exists x P(x)$.

These moves are inference rules, not automatic equivalences. Their universal and existential forms impose different conditions on the object being used.

Rule	Form	Required reading
Universal instantiation	$\forall x P(x) \therefore P(a)$	A universal claim applies to any particular a in the domain.
Universal generalization	$P(a) \therefore \forall x P(x)$	Valid only when a was arbitrary and no premise or temporary assumption restricts a .
Existential instantiation	$\exists x P(x) \therefore P(c)$	Introduce a new name c for one promised object. Do not assume which object it is.
Existential generalization	$P(a) \therefore \exists x P(x)$	One known example establishes existence.

The displayed existential-instantiation line is a reasoning convention, not a standalone equivalence. The name c must be new in the relevant argument, and a conclusion may not depend on a special property of c that was never guaranteed by $P(c)$.

Quantified example

Let the domain be all students. Define

$R(x)$: “ x registered for the exam,” $A(x)$: “ x was assigned a seat,” $L(x)$: “ x arrived late.”

From

$$\forall x (R(x) \rightarrow A(x)) \quad \text{and} \quad \exists x (R(x) \wedge L(x)),$$

prove

$$\exists x (A(x) \wedge L(x)).$$

Read the goal as: “Some student who was assigned a seat arrived late.” The existential premise supplies the student that must be carried through the argument.

Step	Statement	Reason
1	$\exists x (R(x) \wedge L(x))$	premise
2	$R(c) \wedge L(c)$	existential instantiation; c is fresh
3	$R(c)$	simplification, 2
4	$L(c)$	simplification, 2
5	$\forall x (R(x) \rightarrow A(x))$	premise
6	$R(c) \rightarrow A(c)$	universal instantiation, 5
7	$A(c)$	modus ponens, 3 and 6
8	$A(c) \wedge L(c)$	conjunction, 7 and 4
9	$\exists x (A(x) \wedge L(x))$	existential generalization, 8

We did not prove that every late student was assigned a seat. We followed the same student from the existential premise to an existential conclusion.

5 Proof methods

A proof's premises may include definitions, stated hypotheses, axioms, and earlier theorems. An accepted starting statement is an *axiom*; a proved statement is a *theorem*; and an unproved proposal is a *conjecture*. The secondary labels *lemma* and *corollary* only describe how proved results are used.

Many theorems have the form

$$\forall x (P(x) \rightarrow Q(x)).$$

A proof begins by choosing an arbitrary x in the domain and assuming $P(x)$. It must derive $Q(x)$ without using any unannounced special property of x . Because x was arbitrary, the result applies throughout the domain.

Definitions turn the logical goal into usable mathematics. For integers:

$$\begin{aligned} n \text{ is even} &\iff n = 2k \text{ for some } k \in \mathbb{Z}, \\ n \text{ is odd} &\iff n = 2k + 1 \text{ for some } k \in \mathbb{Z}. \end{aligned}$$

To prove that an expression is even, the target is not merely “looks divisible by two.” The target is to write it as 2 times an integer.

5.1 Direct proof

MUST UNDERSTAND **Direct proof.** To prove $P \rightarrow Q$ directly, assume P and derive Q .

Example: the sum of two odd integers

Claim. If m and n are odd integers, then $m + n$ is even.

Proof. Let m and n be arbitrary odd integers. By the definition of odd, there are integers a and b such that

$$m = 2a + 1, \quad n = 2b + 1.$$

Therefore,

$$m + n = (2a + 1) + (2b + 1) = 2(a + b + 1).$$

Because $a + b + 1$ is an integer, $m + n$ is twice an integer. Hence $m + n$ is even. $\square$

The reusable pattern is: unpack the hypotheses, compute, match the definition of the goal, and check every side condition. A conditional is immediate when its premise is impossible (a *vacuous proof*) or its conclusion is already known (a *trivial proof*).

5.2 Proof by contraposition

Because

$$P \rightarrow Q \equiv \neg Q \rightarrow \neg P,$$

we may prove $P \rightarrow Q$ by assuming $\neg Q$ and deriving $\neg P$.

MUST UNDERSTAND **Proof by contraposition.** Assume $\neg Q$ and derive $\neg P$; the original conditional then follows by logical equivalence.

Method	Start by assuming	Aim to derive
Direct proof	P	Q
Contraposition	$\neg Q$	$\neg P$

Contraposition is useful when the negation of the conclusion has a simple mathematical form.

Example: an even square has an even base

Claim. If n^2 is even, then n is even.

Proof by contraposition. Let n be an integer and suppose n is not even. Every integer is either even or odd, so n is odd. Thus $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Because $2k^2 + 2k$ is an integer, n^2 is odd and therefore not even. We have proved that if n is not even, then n^2 is not even. Hence, by contraposition, if n^2 is even, then n is even. $\square$

The direction matters. Proving “ n even implies n^2 even” would prove the converse, not the stated theorem.

5.3 Proof by contradiction

MUST UNDERSTAND **Proof by contradiction.** To prove a statement S by contradiction, assume $\neg S$ and derive an impossibility. For a conditional $P \rightarrow Q$, the negation is

$$\neg(P \rightarrow Q) \equiv P \wedge \neg Q.$$

Thus a contradiction proof of a conditional assumes both the hypothesis and the failure of the conclusion.

Example: $\sqrt{2}$ is irrational

Claim. The number $\sqrt{2}$ is irrational.

Proof by contradiction. Suppose instead that $\sqrt{2}$ is rational. Choose integers a and b with $b \neq 0$ such that

$$\sqrt{2} = \frac{a}{b}$$

and the fraction is in lowest terms. Squaring gives

$$2 = \frac{a^2}{b^2}, \quad \text{so} \quad a^2 = 2b^2.$$

Thus a^2 is even. By the result just proved, a is even, so $a = 2k$ for some integer k . Substituting,

$$(2k)^2 = 2b^2 \implies b^2 = 2k^2.$$

Therefore b^2 is even, so b is even as well. Both a and b are divisible by 2, contradicting the choice of a/b in lowest terms. Hence $\sqrt{2}$ is irrational. $\square$

The contradiction is not simply “ $\sqrt{2}$ is rational and irrational.” It is the concrete conflict between lowest terms and a common factor of 2.

6 Other proof forms

6.1 Biconditional proofs

To prove $P \leftrightarrow Q$, prove both $P \rightarrow Q$ and $Q \rightarrow P$. The two directions may use different methods.

For example, for every integer n ,

$$n \text{ is even} \iff n^2 \text{ is even.}$$

The forward direction is direct: if $n = 2k$, then $n^2 = 2(2k^2)$. The reverse direction is the contraposition result from the preceding section.

To prove several statements $P_1, \dots, P_k$ equivalent, it is enough to prove a cycle

$$P_1 \rightarrow P_2, \quad P_2 \rightarrow P_3, \quad \dots, \quad P_k \rightarrow P_1.$$

Following the arrows lets any statement reach every other statement. A list of unrelated one-way implications is not enough.

6.2 Proof by cases

MUST UNDERSTAND **Proof by cases.** A proof by cases divides the possibilities into exhaustive cases and proves the claim in each one. The case split must cover the entire domain.

Claim. For every integer n , the product $n(n+1)$ is even.

Proof. Let n be an arbitrary integer. There are two cases.

Case 1: n is even. Then $n = 2k$ for some integer k , so

$$n(n+1) = 2(k(n+1)),$$

which is even.

Case 2: n is odd. Then $n = 2k + 1$ for some integer k , so $n + 1 = 2(k + 1)$. Hence

$$n(n + 1) = 2(n(k + 1)),$$

which is even.

Both cases lead to the claim, and every integer is even or odd. Therefore $n(n + 1)$ is even for every integer n . $\square$

An **exhaustive proof** is a proof by cases over a finite list. It is valid only after that list has been shown to cover the domain.

6.3 Existence and uniqueness

To prove $\exists x P(x)$ constructively, give one object a and verify $P(a)$.

Example. There is a positive integer that is both a perfect square and a perfect cube. The number 64 works because

$$64 = 8^2 = 4^3.$$

A **nonconstructive existence proof** guarantees that an object exists without identifying it. In either kind of proof, the claim is only existence; uniqueness is separate.

To prove that exactly one object has a property, prove two parts:

1. **Existence:** at least one object has the property.
2. **Uniqueness:** any two objects with the property are equal.

Example: a unique linear solution

Claim. If $a, b, c \in \mathbb{R}$ and $a \neq 0$, then there is a unique real number x satisfying $ax + b = c$.

Existence. Let

$$x = \frac{c - b}{a}.$$

This is defined because $a \neq 0$, and

$$ax + b = a \left(\frac{c - b}{a} \right) + b = c.$$

Uniqueness. Suppose x_1 and x_2 both satisfy the equation. Then

$$ax_1 + b = c = ax_2 + b,$$

so $a(x_1 - x_2) = 0$. Because $a \neq 0$, we obtain $x_1 = x_2$. Thus the solution exists and is unique. $\square$

Finding one solution proves existence, not uniqueness. Showing that any two solutions coincide proves “at most one,” not existence. Both parts are necessary.

6.4 Counterexamples and proof errors

To disprove a universal claim $\forall x P(x)$, give one object a for which $P(a)$ is false. For example, “every prime is odd” is false because 2 is prime and not odd.

Testing examples can suggest a conjecture, but it does not prove a universal claim over an infinite domain. A finite check becomes a proof only when it exhausts every possible case.

Three common proof errors are structural:

- **Using an invalid implication direction.** From $P \rightarrow Q$ and Q , one cannot infer P .
- **Hiding a failed side condition.** Dividing by an expression requires proving it is nonzero; taking a reciprocal requires the same check.
- **Circular reasoning.** A proof may not use the desired conclusion, or an equivalent statement not yet established, as a premise.

A convincing calculation is not yet a proof until its assumptions, domain, and logical direction are clear.

7 Proof strategy

Start with the logical shape of the claim. It tells you what kind of evidence is required.

Claim shape	Natural first move
$\forall x P(x)$	Choose an arbitrary x and prove $P(x)$.
$\exists x P(x)$	Find or derive one object and verify it.
$P \rightarrow Q$	Assume P for a direct proof, $\neg Q$ for contraposition, or $P \wedge \neg Q$ for contradiction.
$P \leftrightarrow Q$	Prove both directions and label them.
$P_1 \vee \dots \vee P_k$	Prove one disjunct, or derive the disjunction by exhaustive cases.
$\exists! x P(x)$	Separate existence from uniqueness.
Universal claim suspected false	Search for a counterexample.

Then use this workflow:

1. **State the domain, assumptions, and target.**
2. **Expand the relevant definitions.** Turn words such as even, divisible, or unique into the form the proof must produce.
3. **Build a valid route.** Work forward from the hypotheses; use backward reasoning only to discover what would be sufficient for the goal.
4. **Audit the result.** Check quantifier roles, implication directions, exhausted cases, side conditions, and the final target. Change method if the route remains awkward.